THE SUM $\sum_{i=1}^{n} \frac{1}{i}\binom{2n-i-1}{i-1}$ IS NOT AN INTEGER FOR $n \geq 2$

MATH0

Abstract. For $n \geq 1$ let $S(n) = \sum_{i=1}^{n} \frac{1}{i}\binom{2n-i-1}{i-1}$. We prove that $S(n)$ is not an integer for any $n \geq 2$, as conjectured by V. Shevelev and Z. Seidov in 2010 in OEIS A175385. The proof combines the identity $2nS(n)+1 = L_{2n}$, where L_j is the jth Lucas number, with a congruence modulo 4 for even n and, for odd n, an argument in an algebraic closure of $\mathbb{F}_p$, where p is the least prime factor of n. The result is not new: it follows from the classical theorem that no even composite N divides $L_N - 1$ (White, Hunt and Dresel 1977; Di Porto 1993; Bruckman 1994), and a Lean proof found by the AlphaProof Nexus agent was published in 2026, so ours is an independent formal proof of a known result. It is verified in Lean 4 with Mathlib against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the 27 listed terms of the sequence.

1. Introduction

The OEIS sequence A175385 [10], contributed by Vladimir Shevelev and Zak Seidov on April 24, 2010, lists the numerators of the rational numbers

$$S(n) = \sum_{i=1}^{n} \frac{1}{i}\binom{2n-i-1}{i-1}, \qquad n \geq 1,$$

and its companion A175386 [11] lists their denominators: $S(1) = 1$, $S(2) = 3/2$, $S(3) = 17/6$, $S(4) = 23/4$, $S(5) = 61/5$, ... Both entries record the conjecture that $S(n)$ is not an integer for $n > 1$. They also give the closed form $S(n) = \big({}_2F_1(\tfrac{1}{2} - n, -n; 1-2n; -4) - 1\big)/(2n)$.

The problem becomes transparent once one notices the identity

$$2n\,S(n) + 1 = L_{2n} \qquad (n \geq 1), \tag{1}$$

where $L_j = F_{j+1} + F_{j-1}$ are the Lucas numbers [8] and F_j the Fibonacci numbers (Proposition 4 below; it follows from the diagonal sums of Pascal's triangle). Thus $S(n) \in \mathbb{Z}$ if and only if $2n \mid L_{2n} - 1$, and the conjecture says that $2n \nmid L_{2n} - 1$ for every $n \geq 2$.

Prior work. In this form the conjecture is a special case of a classical theorem. A composite number N with $N \mid L_N - 1$ is called a (Bruckman–)Lucas pseudoprime [9], and Bruckman [2] proved that such numbers are odd: no even composite N divides $L_N - 1$. As Bruckman acknowledges in [2], this had been proved earlier by White, Hunt and Dresel [13], and independently by Di Porto [3], who calls such N Fibonacci pseudoprimes of the first kind. For $n \geq 2$ the number $N = 2n \geq 4$ is even and composite, so this theorem and (1) give the conjecture. The conjecture, posed in 2010, is therefore a consequence of results published between 1977 and 1994. This deduction was pointed out by the math0 referees; we do not know whether it was recorded earlier, but it is not mentioned in A175385.

More recently, on May 25, 2026, Ralf Stephan added a comment to A175386 [11] stating that the conjecture was proved by an autonomous AI agent, with references to the paper of Tsoukalas et al. [12] and to a Lean file of Google DeepMind's AlphaProof Nexus [5]; the entry also cites Adamczewski [1, p. 15, Appendix A] for a proof. The Lean file proves, for $n > 1$, that the

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denominator of $S(n)$ is not 1, which is equivalent to the conjecture. It also starts from (1) and splits on the parity of n. For odd n it uses $L_{2n} = L_n^2 + 2$, the factorization $F_{3n} = F_n(L_n^2 + 1)$, the divisibility $p \mid F_{p^2-1}$ for primes $p \neq 2, 5$, and the strong divisibility property of the Fibonacci numbers to conclude that the least prime factor of n divides $F_3 = 2$. The entry A175385 itself (revision #14, March 2023) records no proof.

This paper. We give a self-contained proof of the conjecture and an independent Lean formalization of it, checked against a statement that math0 formalized from A175385 (Section 2). The case of even n is a congruence modulo 4, as in [5]. For odd n we argue in an algebraic closure of $\mathbb{F}_p$, where p is the least prime factor of n: a root α of $X^2 - X - 1$ satisfies $\alpha^{p^2-1} = 1$, and from $L_{2n} \equiv 1 \pmod{p}$ we derive $\alpha^6 = -1$, which forces $p \mid 20$. Arguments of this kind are standard in the study of Lucas sequences modulo primes, and we claim no novelty for the mathematics. We classify the result as an *independent formal proof of a known result*.

2. Statement

Theorem 1. *For every integer $n \geq 2$, the rational number $S(n) = \sum_{i=1}^{n} \frac{1}{i}\binom{2n-i-1}{i-1}$ is not an integer.*

The COMMENTS section of A175385 [10] reads, verbatim:

> `We conjecture that Sum_{i=1..n} ((1/i)*C(2n-i-1,i-1)) is not an integer for n>1.`

The entry is authored by Vladimir Shevelev and Zak Seidov and dated April 24, 2010; the name of the sequence is `a(n) = numerator of Sum_{i=1..n} binomial(2n-i-1,i-1)/i` (offset 1). The same conjecture appears in A175386 [11].

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A175385.lean`; it was reviewed for fidelity and frozen (SHA-256 `3e8f66ae...c8654`) before any proof attempt. After the copyright header, the import of Mathlib and the module docstring, the file reads verbatim:

```
namespace Math0.Statements.A175385

/-- The sum `∑_{i=1}^{n} C(2n-i-1, i-1)/i`, computed exactly in `ℚ`. -/
def S (n : ℕ) : ℚ :=
  ∑ i ∈ Finset.Icc 1 n, ((Nat.choose (2 * n - i - 1) (i - 1) : ℕ) : ℚ) / (i : ℚ)

/-- A175385 (offset `1`): the numerator, in lowest terms, of
`∑_{i=1..n} binomial(2n-i-1, i-1)/i`. -/
def a (n : ℕ) : ℕ := (S n).num.natAbs

/-- The definition reproduces all 27 terms listed in OEIS, `a 1, ..., a 27` (kernel-checked). -/
theorem a_data : (List.range 27).map (fun i => a (i + 1)) =
    [1, 3, 17, 23, 61, 107, 421, 1103, 5777, 7563, 19801, 103681, 135721, 355323, 1860497,
      2435423, 6376021, 11128427, 43701901, 114413063, 599074577, 784198803, 2053059121,
      10749957121, 14071876561, 36840651123, 192900153617] := by
  decide +kernel

/-- Conjecture (Shevelev and Seidov, A175385): for every `n > 1`, the sum
`∑_{i=1..n} C(2n-i-1, i-1)/i` is not an integer. -/
def statement : Prop :=
  ∀ n : ℕ, 1 < n → ¬ ∃ k : ℤ, S n = (k : ℚ)

end Math0.Statements.A175385
```

The definition `S` is the defining sum of the entry, computed exactly in $\mathbb{Q}$ with Mathlib's `Nat.choose`; it is not the hypergeometric formula. Subtraction in $\mathbb{N}$ is truncated, but for $1 \le i \le n$ we have $2n - i - 1 \ge n - 1 \ge 0$ and $i - 1 \ge 0$, so no truncation occurs, and the divisor i is at least 1. The function `a` is the numerator of $S(n)$ in lowest terms (`Rat.num`, converted to $\mathbb{N}$ by `Int.natAbs`, which is harmless since $S(n) > 0$ for $n \ge 1$), with the OEIS offset 1. The lemma `a_data` checks, by `decide +kernel`, that $a(1), \dots, a(27)$ are exactly the 27 terms listed in the entry; the Lean kernel evaluates the 27 rational sums from the definitions and compares the numerators. This ties `S` to the OEIS data; `a` itself does not occur in `statement`. Finally, "is not an integer" is encoded as the nonexistence of $k \in \mathbb{Z}$ with $S(n) = k$ in $\mathbb{Q}$, and "for $n > 1$" as the hypothesis `1 < n` on $n \in \mathbb{N}$. The excluded value $n = 1$ is a genuine exception, since $S(1) = 1$.

3. Proof

We write $F_0 = 0$, $F_1 = 1$, $F_{j+2} = F_{j+1} + F_j$ (Mathlib's `Nat.fib`), and use the convention $\binom{a}{b} = 0$ for $b > a$. For $n \ge 1$ we put

$$T(n) = F_{2n+1} + F_{2n-1},$$

which is the Lucas number L_{2n}; the formal proof works with $T(n)$ directly and never defines Lucas numbers. The proof follows the structure of the Lean file; Section 4 lists the corresponding declarations.

3.1. The closed form.

Lemma 2 (Diagonal sums of Pascal's triangle)**.** *If $m, N \in \mathbb{N}$ and $m < 2N$, then $F_{m+1} = \sum_{k=0}^{N-1} \binom{m-k}{k}$.*

Proof. Mathlib's `Nat.fib_succ_eq_sum_choose` states $F_{m+1} = \sum_{a+b=m} \binom{a}{b}$; reindexing the antidiagonal gives $F_{m+1} = \sum_{k=0}^{m} \binom{m-k}{k}$. If $k \ge N$ then $2k \ge 2N > m$, so $m - k < k$ (in truncated subtraction) and $\binom{m-k}{k} = 0$; likewise for $k > m$. Hence both $\sum_{k=0}^{m}$ and $\sum_{k=0}^{N-1}$ equal the sum over $0 \le k \le N + m$. □

Lemma 3. *For $1 \le i \le n$,*

$$\frac{2n}{i}\binom{2n-i-1}{i-1} = \binom{2n-i}{i} + \binom{2n-i-1}{i-1}.$$

Proof. The identity $(M+1)\binom{M}{k} = (k+1)\binom{M+1}{k+1}$ (`Nat.add_one_mul_choose_eq`) with $M = 2n - i - 1$ and $k = i - 1$ gives $(2n-i)\binom{2n-i-1}{i-1} = i\binom{2n-i}{i}$. Adding $i\binom{2n-i-1}{i-1}$ to both sides and dividing by i gives the claim. □

Proposition 4. *For every $n \ge 1$, $2n\,S(n) + 1 = T(n) = F_{2n+1} + F_{2n-1}$.*

Proof. Write $n = m+1$ and substitute $i = j+1$; then $S(n) = \sum_{j=0}^{m} \binom{2m-j}{j}/(j+1)$. By Lemma 3,

$$2n\,S(n) = \sum_{j=0}^{m} \binom{2m+1-j}{j+1} + \sum_{j=0}^{m} \binom{2m-j}{j}.$$

Lemma 2 with (m, N) replaced by $(2m+2, m+2)$ gives $F_{2m+3} = \sum_{k=0}^{m+1} \binom{2m+2-k}{k} = 1 + \sum_{j=0}^{m} \binom{2m+1-j}{j+1}$, where we split off $k = 0$ and put $k = j+1$. Lemma 2 with $(2m, m+1)$ gives $F_{2m+1} = \sum_{j=0}^{m} \binom{2m-j}{j}$. Hence $2nS(n) = (F_{2m+3} - 1) + F_{2m+1} = T(n) - 1$. □

3.2. **Lucas numbers in a commutative ring.**

Lemma 5. *Let R be a commutative ring and $\alpha \in R$ with $\alpha^2 = \alpha + 1$. Then:*

(1) $\alpha^{j+1} = F_{j+1}\alpha + F_j$ *for all* $j \geq 0$*;*
(2) $\beta = 1 - \alpha$ *satisfies* $\beta^2 = \beta + 1$ *and* $\alpha\beta = -1$*;*
(3) $\alpha^{2n} + (1-\alpha)^{2n} = T(n) \cdot 1_R$ *for all* $n \geq 1$.

Proof. (1) By induction on j: the case $j = 0$ is trivial, and $\alpha^{j+2} = \alpha(F_{j+1}\alpha + F_j) = F_{j+1}(\alpha + 1) + F_j\alpha = F_{j+2}\alpha + F_{j+1}$. (2) $\beta^2 = 1 - 2\alpha + \alpha^2 = 2 - \alpha = \beta + 1$ and $\alpha\beta = \alpha - \alpha^2 = -1$. (3) Apply (1) to α and to β with $j = 2n - 1$ and add: $\alpha^{2n} + \beta^{2n} = F_{2n}(\alpha + \beta) + 2F_{2n-1} = F_{2n} + 2F_{2n-1} = F_{2n+1} + F_{2n-1}$. □

3.3. **Even n.**

Lemma 6. $F_{j+6} \equiv F_j \pmod 4$ *for all* $j \geq 0$. *Consequently, if* $n \geq 2$ *is even, then* $T(n) \equiv 2$ *or* $3 \pmod 4$.

Proof. The addition formula $F_{a+b+1} = F_aF_b + F_{a+1}F_{b+1}$ (`Nat.fib_add`) with $a = 5$ gives $F_{j+6} = 5F_j + 8F_{j+1} \equiv F_j \pmod 4$. So $F_j \bmod 4$ depends only on $j \bmod 6$, and $(F_0, \ldots, F_5) \equiv (0, 1, 1, 2, 3, 1) \pmod 4$. Write $n = m + 1$ with m odd, so $m \equiv 1, 3, 5 \pmod 6$ and correspondingly $(2m + 3, 2m + 1) \equiv (5, 3), (3, 1), (1, 5) \pmod 6$. Hence $T(n) = F_{2m+3} + F_{2m+1} \equiv 1 + 2,\ 2 + 1,\ 1 + 1$, that is $3, 3, 2 \pmod 4$. □

3.4. **Odd n.**

Lemma 7. *Let* $n \geq 1$ *and suppose that* $S(n) = k$ *with* $k \in \mathbb{Z}$. *Then* $T(n) = 2nk + 1$ *in* $\mathbb{Z}$, *and* $T(n) \cdot 1_R = 1$ *in every commutative ring* R *whose characteristic divides* n.

Proof. The first claim is Proposition 4, transported from $\mathbb{Q}$ to $\mathbb{Z}$ by injectivity of $\mathbb{Z} \to \mathbb{Q}$. Mapping it to R, where $n \cdot 1_R = 0$, gives the second. □

Lemma 8 (Frobenius)**.** *Let K be a field of prime characteristic p and $\alpha \in K$ with $\alpha^2 = \alpha + 1$. Then $\alpha \neq 0$ and $\alpha^{p^2-1} = 1$.*

Proof. Since $x \mapsto x^p$ is a ring endomorphism of K, $y = \alpha^p$ satisfies $y^2 = (\alpha^2)^p = (\alpha+1)^p = y+1$. Then $(y-\alpha)\big(y-(1-\alpha)\big) = y^2 - y + \alpha - \alpha^2 = (y+1) - y + \alpha - (\alpha+1) = 0$, so $y = \alpha$ or $y = 1-\alpha$. In the first case $\alpha^{p^2} = y^p = \alpha^p = \alpha$; in the second $\alpha^{p^2} = (1-\alpha)^p = 1 - \alpha^p = 1 - (1-\alpha) = \alpha$. Finally $\alpha \neq 0$ because $0^2 \neq 0 + 1$, and cancelling α from $\alpha^{p^2} = \alpha$ gives $\alpha^{p^2-1} = 1$. □

Lemma 9. *Let* $n > 1$ *be odd and let* p *be its least prime factor. Then* $\gcd(n, p^2 - 1) = 1$.

Proof. Write $p = r + 1$, so $r \geq 1$, r is even (as p is odd), and $p^2 - 1 = r(r + 2)$. Let q be a prime dividing both n and $r(r + 2)$. Then $q \geq p = r + 1$, and q is odd because n is. If $q \mid r$ then $q \leq r$, which is impossible. If $q \mid r + 2$ then $q \leq r + 2$, so $q = r + 1$ or $q = r + 2$. In the first case $r + 1$ divides $(r + 2) - (r + 1) = 1$, so $r = 0$, a contradiction; in the second $q = r + 2$ is even, a contradiction. □

Lemma 10. *Let K be a field of prime characteristic $p \notin \{2, 5\}$, let $\alpha \in K$ with $\alpha^2 = \alpha + 1$, and let N be an odd natural number with* $\gcd(N, p^2 - 1) = 1$. *Then* $\alpha^{2N} + (1-\alpha)^{2N} \neq 1$.

Proof. Suppose $\alpha^{2N} + (1 - \alpha)^{2N} = 1$ and put $x = \alpha^{2N}$, $x' = (1 - \alpha)^{2N}$. By Lemma 5(2), $xx' = (\alpha(1 - \alpha))^{2N} = (-1)^{2N} = 1$. Hence

$$x^2 - x + 1 = x(x + x' - 1) - (xx' - 1) = 0,$$

and $x^3 + 1 = (x + 1)(x^2 - x + 1) = 0$, that is $\alpha^{6N} = -1$ and $\alpha^{12N} = 1$. Let d be the multiplicative order of α, which is finite by Lemma 8. Then $d \mid 12N$, and $d \mid p^2 - 1$ by

Lemma 8; since $\gcd(N, p^2-1) = 1$ we get $\gcd(d, N) = 1$ and hence $d \mid 12$, i.e. $\alpha^{12} = 1$. Writing $N = 2t+1$, we obtain $-1 = \alpha^{6N} = (\alpha^{12})^t \alpha^6 = \alpha^6$. By Lemma 5(1), $\alpha^6 = F_6\alpha + F_5 = 8\alpha + 5$, so $8\alpha = -6$. Using $\alpha^2 = \alpha + 1$ we compute $64\alpha^2$ in two ways: $64\alpha^2 = (8\alpha)^2 = 36$ and $64\alpha^2 = 8\cdot(8\alpha)+64 = -48+64 = 16$. Hence $20 = 0$ in K, so $p \mid 20$ and $p \in \{2, 5\}$, a contradiction. (The Lean proof obtains $20 = 0$ in one step from the identity $20 = (14 - 8X)(X^6+1) + (8X^5 - 6X^4 + 2X^3 - 4X^2 - 2X - 6)(X^2 - X - 1)$ in $\mathbb{Z}[X]$, evaluated at $X = \alpha$.) □

3.5. **Proof of Theorem 1.** Let $n \geq 2$ and suppose that $S(n) = k \in \mathbb{Z}$. By Lemma 7, $T(n) = 2nk + 1$.

Case n even. Write $n = 2t$. Then $T(n) = 4tk + 1 \equiv 1 \pmod 4$, contradicting Lemma 6.

Case n odd. Let p be the least prime factor of n; then $p \neq 2$.

Subcase $p = 5$. Take $R = \mathbb{Z}/5\mathbb{Z}$, whose characteristic 5 divides n, so $T(n) = 1$ in $\mathbb{Z}/5\mathbb{Z}$ by Lemma 7. The element $\alpha = 3$ satisfies $\alpha^2 = 9 = 4 = \alpha + 1$ in $\mathbb{Z}/5\mathbb{Z}$, and $1 - \alpha = -2 = 3$. By Lemma 5(3) and since $3^2 = -1$ and n is odd, $T(n) = 3^{2n} + 3^{2n} = 2\,(3^2)^n = 2\,(-1)^n = -2 = 3$ in $\mathbb{Z}/5\mathbb{Z}$. As $3 \neq 1$ in $\mathbb{Z}/5\mathbb{Z}$, this is a contradiction.

Subcase $p \neq 5$. Let K be an algebraic closure of $\mathbb{Z}/p\mathbb{Z}$; it is a field of characteristic p. Choose $s \in K$ with $s^2 = 5$ (Mathlib's `IsAlgClosed.exists_eq_mul_self`); since $p \neq 2$, $2 \neq 0$ in K, and $\alpha = (1+s)/2$ satisfies $\alpha^2 = (6+2s)/4 = \alpha + 1$. By Lemma 5(3) and Lemma 7 applied to $R = K$, $\alpha^{2n} + (1-\alpha)^{2n} = T(n) \cdot 1_K = 1$. But n is odd and $\gcd(n, p^2-1) = 1$ by Lemma 9, so Lemma 10 with $N = n$ shows that $\alpha^{2n} + (1-\alpha)^{2n} \neq 1$, a contradiction.

In all cases the assumption $S(n) \in \mathbb{Z}$ is contradictory, which proves Theorem 1. □

Remark 11. The prime $p = 3$ needs no separate treatment: it is covered by the last subcase. The prime $p = 5$ must be treated separately, because $X^2 - X - 1 = (X-3)^2$ over $\mathbb{F}_5$ and $20 = 0$ in characteristic 5, so Lemma 10 gives nothing. The algebraic closure is used only to provide a square root of 5; the field $\mathbb{F}_{p^2}$ would suffice. By (1), Theorem 1 is equivalent to the assertion that no even $N \geq 4$ divides $L_N - 1$, which is exactly the theorem that Lucas pseudoprimes are odd [13, 3, 2], since the even composite numbers are the even numbers $N \geq 4$. So the argument above is also a proof of that theorem; we have not compared it with the published proofs.

4. Machine verification

Files and declarations. The proof is the file `lean/Math0/Frontier/OEIS/A175385.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file of Section 2 and the audit commands of `lean/Math0/Audit.lean`. Its only public declaration is `theorem result : statement` in the namespace `Math0.Frontier.A175385`, so `Math0.Frontier.A175385.result` has type exactly `Math0.Statements.A175385.statement`. All auxiliary lemmas are `private`. They correspond to Section 3 as follows:

Lemma 2	`fib_succ_eq_sum_range`, `fib_succ_eq_sum_range'`
Lemma 3	`term_identity`
Proposition 4	`S_succ`, `main_identity`
Lemma 5	`pow_eq_fib`, `lucas`
Lemma 6	`fib_add_six_mod`, `fib_mod_four`, `T_mod_four`
Lemma 8	`frob`
Lemma 9	`coprime_minFac`
Lemma 10	`field_part`, `prime_dvd_twenty`
Lemma 7, Section 3.5	`result` (with the local fact `key`)

The least prime factor is Mathlib's `Nat.minFac`, the characteristic is expressed by the class `CharP`, the algebraic closure is `AlgebraicClosure (ZMod p)`, and the multiplicative order is `orderOf`. The Frobenius step uses `add_pow_char` and `sub_pow_char` directly, with no appeal to the cardinality of a finite field. The finite computations are all small: $F_5 = 5$ and $F_6 = 8$ in

`fib_add_six_mod` (by `rfl`); the residues $F_j \bmod 4$ for $j \le 5$ in `T_mod_four`, and $3^2 = 3 + 1$, $1 - 3 = 3$, $3^2 = -1$ and $-1 + (-1) \neq 1$ in $\mathbb{Z}/5\mathbb{Z}$ (by `decide`); and the primes dividing 20 in `prime_dvd_twenty` (by `interval_cases`). The file uses no `sorry`, `native_decide`, `set_option`, new axioms or new syntax.

Audit. The file ends with the command `#math0_settles`, applied to `Math0.Frontier.A175385.result` and `Math0.Statements.A175385.statement`; this command is defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself. It fails unless (1) `statement` is a definition of type `Prop` without universe parameters; (2) `statement` and `result` depend on no axioms other than `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, the axiom `Lean.ofReduceBool` introduced by `native_decide`, and custom axioms); and (3) the type of `result` is syntactically the constant `statement` or its negation, after removing metadata annotations and without unfolding. On this file it reports

```
math0 audit: 'Math0.Statements.A175385.statement' depends on axioms [Classical.choice, Quot.
    sound, propext]
math0 audit: 'Math0.Frontier.A175385.result' depends on axioms [Classical.choice, Quot.sound,
    propext]
math0 statement check: 'Math0.Frontier.A175385.result' proves 'Math0.Statements.A175385.
    statement'
```

The SHA-256 hash of the statement file is unchanged from the value recorded when it was frozen.

Toolchain and reproduction. We use Lean 4 [7] `v4.33.1` and Mathlib [6] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [4] at commit `c77fdd4`, which this result does not use, since the statement is not part of formal-conjectures. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A175385.lean` in the directory `lean`; this takes a few minutes on our machine, most of it spent loading Mathlib.

References

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Produced by Claude (an AI model developed by Anthropic) in the math0 project